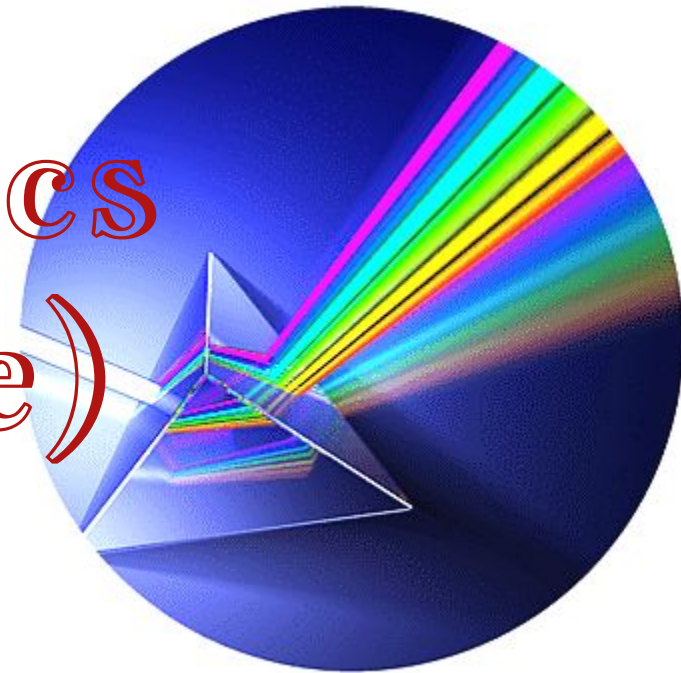




Physical Optics (Interference)



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Newton's Corpuscular Theory

Physical optics is the study of light as a wave phenomenon. Some phenomena such as interference and diffraction can only be explained by treating light as a wave.

The 'Corpuscular theory of light was proposed by Newton in 1675. In this theory, he successfully explained the particle nature of light. The corpuscular theory is the simplest theory of light in which light is assumed as the tiny particles called 'corpuscles'. The corpuscular theory is often referred to as particle theory or Newton's theory of light.

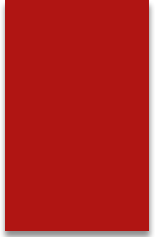
Newton's Corpuscular Theory Statement

According to the Newton's corpuscular theory:

- Light is made up of tiny particles called 'corpuscles' having negligible mass.
- These particles (corpuscles) are perfectly elastic.
- The corpuscles are emitted from luminous sources such as the Sun, candle, electric lamp etc.
- The tiny particles (corpuscles) always travel in a straight line in all directions.
- Each particle (corpuscle) carries kinetic energy with it while moving.
- The corpuscles travel at high velocity.
- The corpuscles (light) would travel faster in the denser medium than in rarer medium. But later this is proved wrong. We know that light travels faster in the rarer medium than in the denser medium.
- When the particles (corpuscles) fall on the retina of the eye, they produce an image of the object or sensation of vision.
- The corpuscles can be of different sizes. The different colors of light are due to the different sizes of the corpuscles.

Explained-----

- Rectilinear propagations of light
- Reflection
- Refraction



Failures of Newton's corpuscular theory

- Newton assumed that light travels faster in the denser medium than in the rarer medium which has been proved wrong.
- The assumption that the different colors of light are due to the differences in the sizes of corpuscles has no justification.
- Newton assumed that reflection of light is due to the repulsion between the corpuscles and the reflecting surface while refraction of light is due to the attraction between the corpuscles and refracting surface. Thus, a medium could either attract or repel the corpuscles of light. But later this statement was proved wrong. We all know that glass partially reflects and partially refracts light. How could glass attract and repel the corpuscles of light simultaneously.
- Corpuscular theory cannot explain the phenomena of diffraction, interference, and polarization of light.

What is Huygens' theory of light?

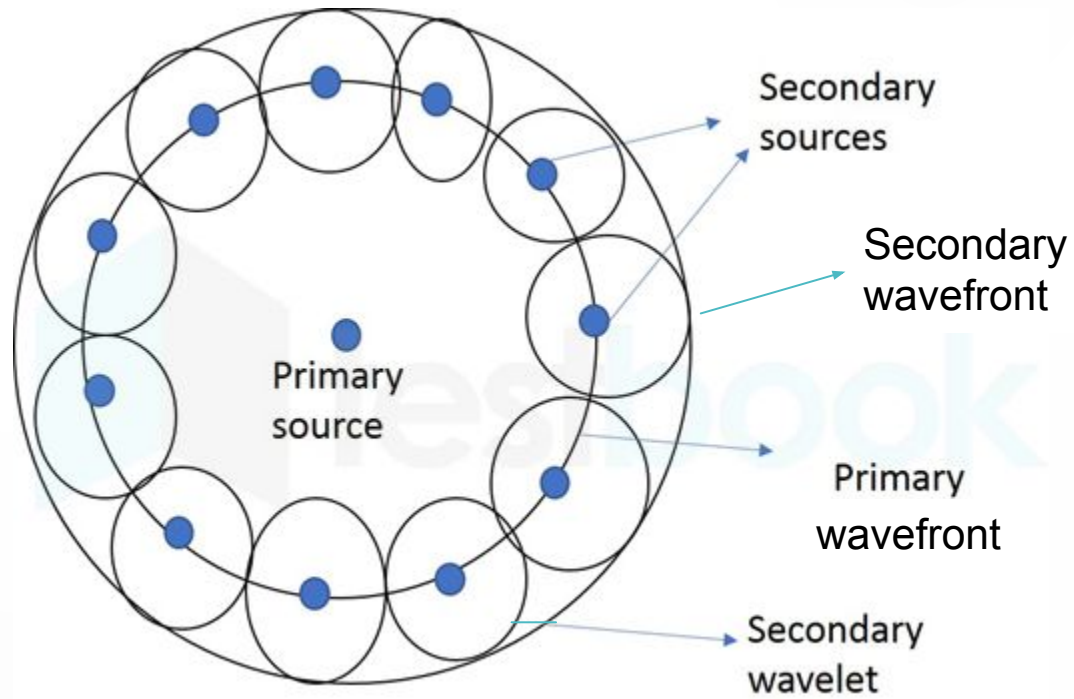
Christiaan Huygens proposed the Huygens principle. In 1678, he changed the way we think about light and its properties. One of the most important ways for examining various optical phenomena is the Huygens principle. Christiaan Huygens proved that his **wave theory of light could explain the phenomena of diffraction, interference, and reflection**. Huygens' theory states that each point in a source of light sends a **wavefront** in all directions in a continuous and homogeneous medium called an ether.

Below are some of the important considerations of the Huygens' principle.

1. A **wavefront** is a locus of all the points of medium to which waves arrive simultaneously so that all the points are in the same phase
2. All the rays are always perpendicular to the wavefront.

3. **Types of wavefronts** – There are three types of wavefronts, that are

- 1. Spherical wavefront:** Consider the 'S' point source of light. Light waves generated by S can move in any direction. If c is the speed of light, then each wave will reach the surface of a sphere with radius ct and centre S after time t . This spherical surface is known as the spherical wavefront at time t , and it is derived from a finite-distance point source. For example, consider the light emitted by a bulb at a limited distance.
- 2. Plane wavefront:** The spherical wavefront is so enormous at a very vast distance from the point source that a tiny portion of it may be considered a plane wavefront. It is derived from an infinitely distant point source. As an example, consider the wavefront caused by sunlight.
- 3. Cylindrical wavefront:** If the source of light is linear i.e., a slit, then we get a cylindrical wavefront. It is obtained from a linear source kept at a finite distance. For example, light is emitted by a fluorescent tube.
- 4. Each point on a wavefront acts like a source of secondary disturbances which emit their own wavelets called secondary wavelets which travel in all direction with speed of wave.**



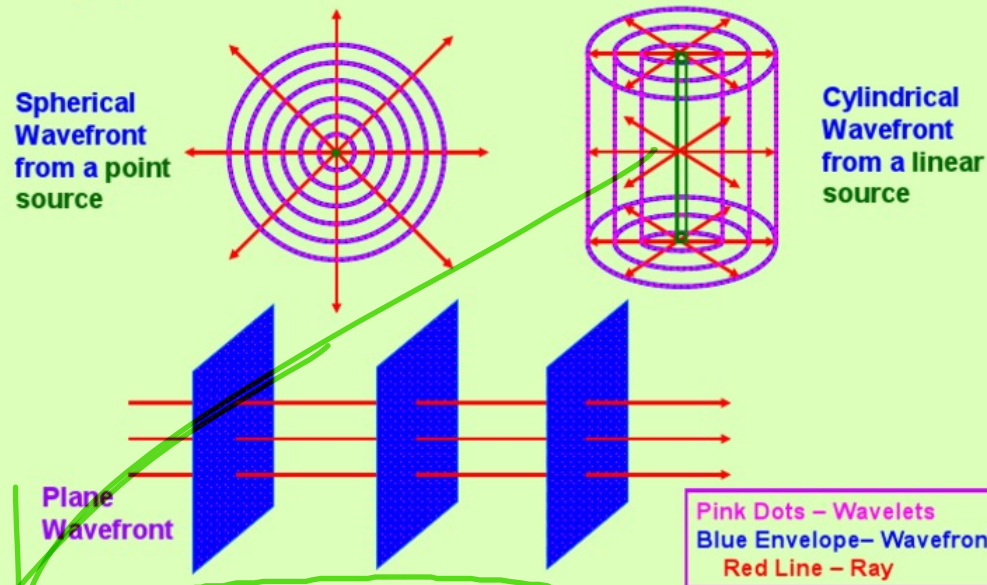
5. Common tangent on the secondary wavelets will give the new wavefront

Wavefront:

A wavelet is the point of disturbance due to propagation of light.

A wavefront is the locus of points (wavelets) having the same phase of oscillations.

A line perpendicular to a wavefront is called a 'ray'.

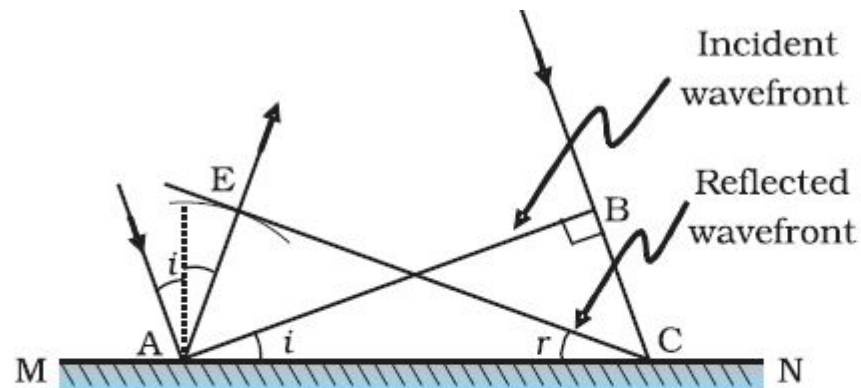


Huygens' Construction or Huygens' Principle of Secondary Wavelets:



1. Each point on a wavefront acts as a fresh source of disturbance of light.
2. The new wavefront at any time later is obtained by taking the forward

Proof of laws of reflection and refraction by Huygens' principle



Reflection of a plane wave AB by the reflecting surface MN.
AB and CE represent incident and reflected wavefronts.

Let v_1 and v_2 represent the speed of light in medium 1 and medium 2 respectively.
 Let wavefront AB be incident on the interface at an angle i .
 If τ is the time taken by the wavefront to travel distance BC, i.e. $BC = v_1\tau$
 Draw a sphere of radius $v_2\tau$ from point A in the second medium.
 CE is the tangent plane drawn from point C on to the sphere. CE is the refracted wavefront.
 $AE = v_2\tau$

Consider $\triangle ABC$ and $\triangle AEC$,

$$\sin i = \frac{BC}{AC} = \frac{v_1\tau}{AC}$$

$$\sin r = \frac{AE}{AC} = \frac{v_2\tau}{AC}$$

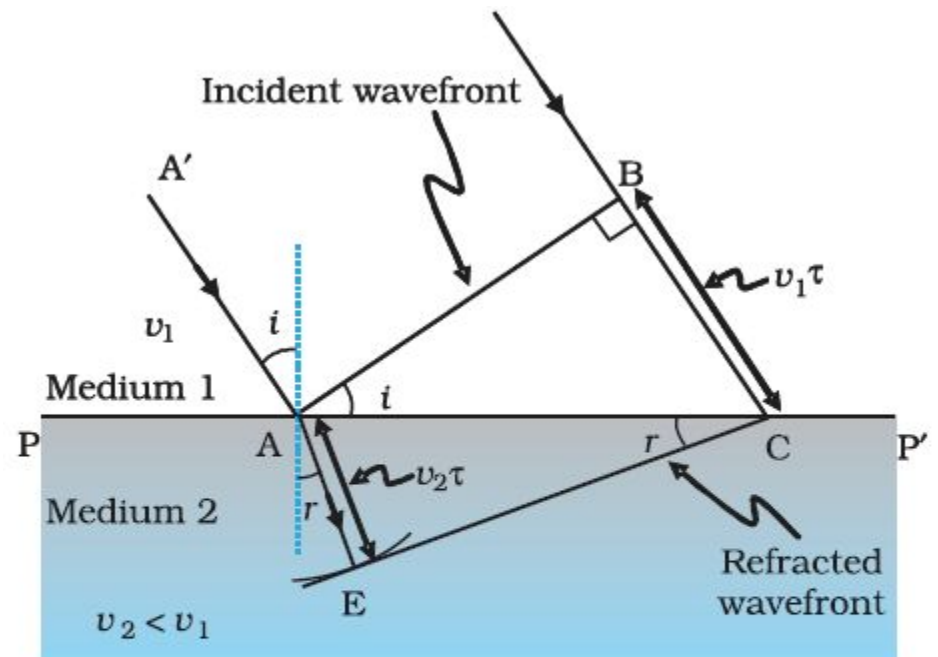
$$\Rightarrow \frac{\sin i}{\sin r} = \frac{v_1}{v_2}$$

$$\text{For medium 1, } n_1 = \frac{c}{v_1}$$

$$\text{For medium 2, } n_2 = \frac{c}{v_2}$$

$$\Rightarrow n_1 \sin i = n_2 \sin r$$

This is Snell's law of refraction.



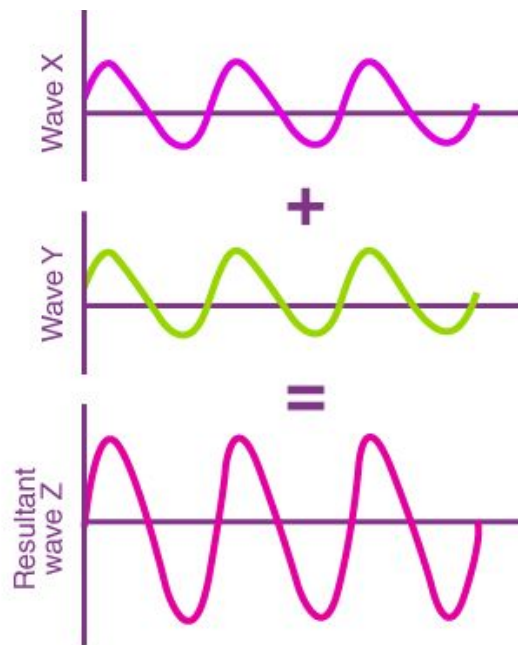
What is Interference?

Interference of waves is the phenomenon when two waves meet or superimpose each other while traveling along the same medium. This resultant interference of two or more waves causes the medium to take a new shape that results from the net effect of the two individual waves upon the particles of the medium. Thus we can say that when two waves meet or come together, the result is the sum of the individual waves.

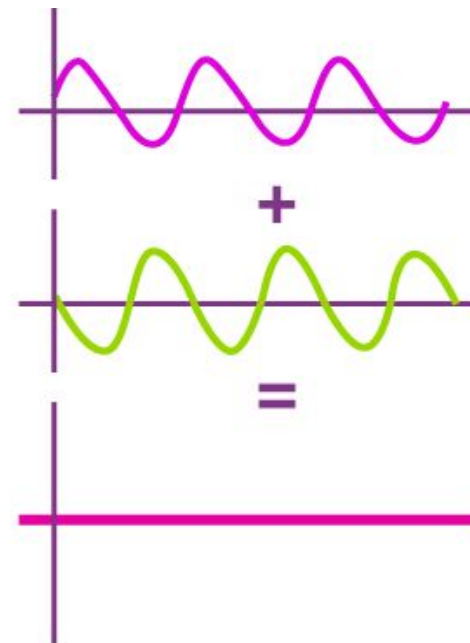
You may wonder what will happen when two waves travelling in the same medium are meeting each other? Will there be any change in frequency of resultant wave or will there be change in amplitude? Also questions may arise will there be any change in the nature of two waves meeting?

To answer all these questions we need to understand the **phenomenon of superposition** and interference of two waves.

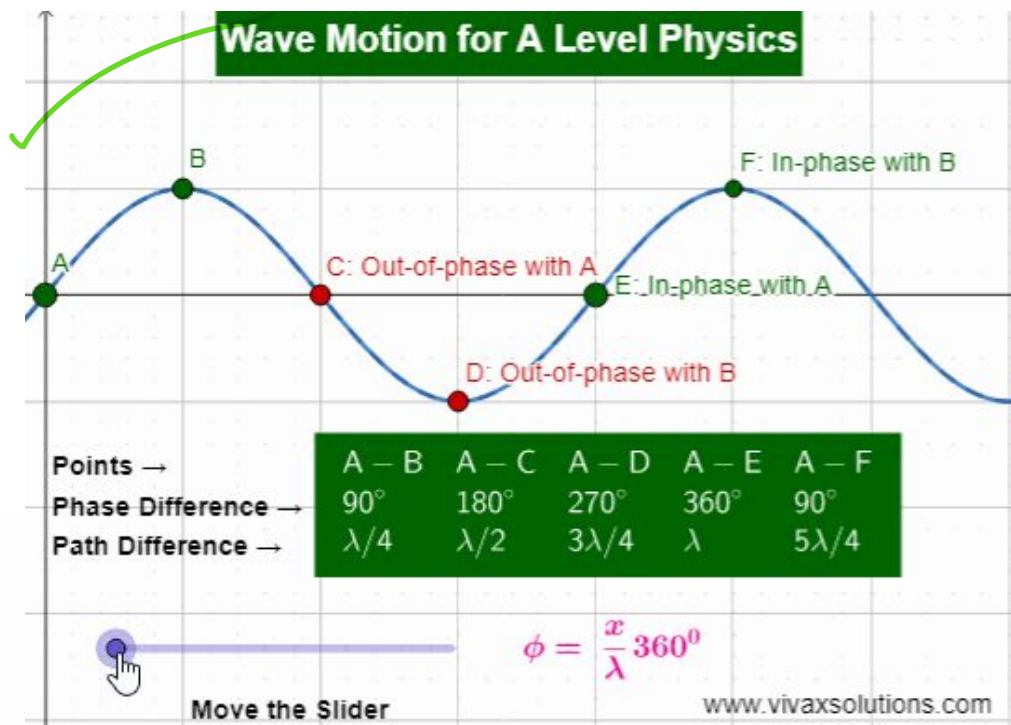
The principle of superposition of waves states that when two or more waves of the same type are incident on the same point, the resultant amplitude at that point is equal to the vector sum of the amplitudes of the individual waves. If the crest of a wave meets the crest of another wave of the same frequency at the same point, then the resultant amplitude is the sum of individual amplitudes – this is known as **constructive interference**. Similarly, suppose a wave's crest meets another wave's trough. In that case, the resultant amplitude is equal to the difference in the individual amplitudes – this is known as **destructive interference**.



Constructive interference



Destructive interference



The phase of a point implies its direction of vibration on a wave. For example, both points A and E vibrate exactly the same way are said to be in phase: when A goes up so does E; the phase difference is either 0° or 360° .

If you consider the motion of points A and C, on the other hand, when A goes down C goes up or vice versa. Therefore, A and C are said to be out of phase; the phase difference is 180° .

That means the phase difference between two points on a wave - or two waves for that matter - can take any value between 0° and 360° .

The phase difference between points A and B, for instance, is 90° .

From the above examples, it is clear there is a connection between the path difference between two points - or two waves - and phase difference. It is as follows:

Phase difference = (path difference / wavelength) x 360

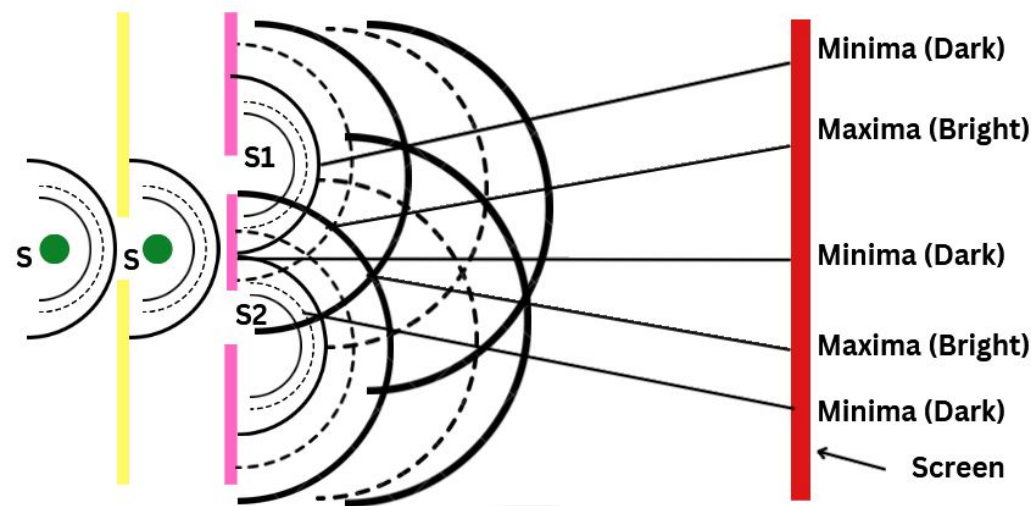
$\phi = x/\lambda \times 360^\circ$, where x is the distance between two points or waves.

Young's double slit experiment

Young placed a monochromatic source (S) of light in front of a narrow slit S_0 and arranged two very narrow slits S_1 and S_2 close to each other in front of slit S_0 . Slits S_1 and S_2 are equidistant from S_0 , so the spherical wavefronts emitted by slit S_0 reach the slits S_1 and S_2 in equal time.

These wavefronts after arriving at S_1 and S_2 spread out of these slits. Thus the emerging waves are of the same amplitude and wavelength and are in phase. Hence slits S_1 and S_2 behave as coherent sources.

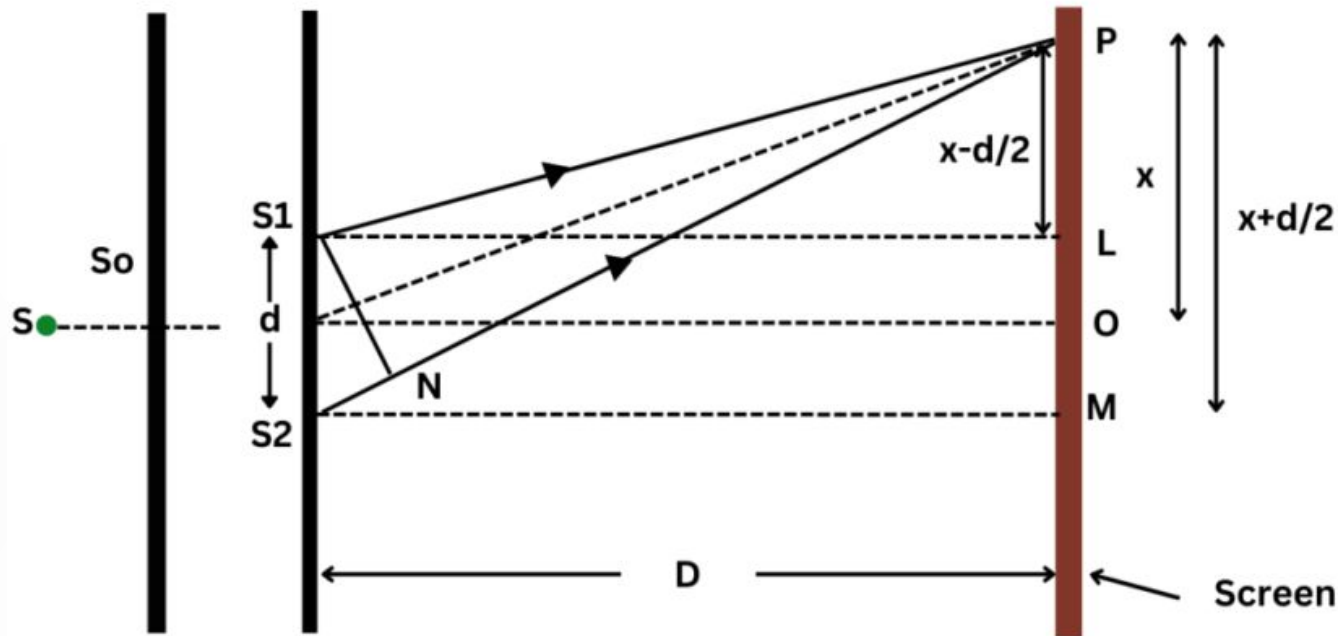
The wavefronts emitted by coherent sources S_1 and S_2 superpose and give rise to interference. When these wavefronts are received on the screen, interference fringes are seen as shown in **Young's double-slit experiment diagram** below.



The points where the destructive interference takes place, we get minima or dark fringe and where the constructive interference takes place, maxima or bright fringe is obtained. The pattern of these dark and bright fringes obtained on the screen is called interference pattern.

Theory of the Experiment

Suppose S is the monochromatic source of light. S_0 is the slit through which the light passes and illuminates the slits S_1 and S_2 . The waves emitted by slits S_1 and S_2 are part of the same wavefront, so these waves have the same frequency and the same phase.



Hence slits S_1 and S_2 behave as two coherent sources. Interference takes place on the screen. If we consider a point O on the perpendicular bisector of S_1S_2 , the waves traveling along S_1O and S_2O have traveled equal distances. Hence they will arrive at O in phase and interfere constructively to make O the centre of a bright fringe or maxima.

✓ To locate the position of the maxima and minima on both sides of O, consider any point P at a distance x from O. Join S_1P and S_2P . Now draw S_1N normal on S_2P . Then the path difference between S_2P and S_1P

$$S_2N = S_2P - S_1P$$

Now from $\triangle S_1PL$,

$$(S_1P)^2 = (S_1L)^2 + (LP)^2 = D^2 + \left(x - \frac{d}{2}\right)^2$$

and from $\triangle S_2PM$,

$$(S_2P)^2 = (S_2M)^2 + (MP)^2$$

$$(S_2P)^2 - (S_1P)^2 = (D)^2 + \left(x + \frac{d}{2}\right)^2 - D^2 - \left(x - \frac{d}{2}\right)^2 = 2xd$$

$$(S_2P - S_1P)(S_2P + S_1P) = 2xd$$

$$(S_2P - S_1P) = \frac{2xd}{(S_2P + S_1P)}$$

Since the distance of screen from slits S_1 and S_2 is very large, so $S_2P \approx S_1P \approx D$

Path difference,

$$(S_2P - S_1P) = \frac{2xd}{2D} = \frac{xd}{D}$$

Maxima or Bright fringes

If the path difference ($S_2P - S_1P$) = xd/D is an integral multiple of λ , then the point P will be the position of the bright fringe or maxima.

That is for bright fringe,

$$\frac{xd}{D} = p\lambda$$
$$x = \frac{p\lambda D}{d} \dots\dots\dots(1)$$

Eqn. (1) gives the position of different bright fringes.

IF

$P = 0$, $x = 0$, i.e., the central fringe at O will be bright.

IF

$$p = 1, \quad x_1 = \frac{\lambda D}{d}$$

This is the position of first bright fringe w.r.t. point O.

IF

$$p = 2, \quad x_2 = \frac{2\lambda D}{d}$$

This is the position of second bright fringe w.r.t. point O.

IF
$$p = p, \quad x_p = \frac{p\lambda D}{d}$$

This is the position of **pth** bright fringe w.r.t. point O.

IF
$$p = (p + 1), \quad x_{p+1} = \frac{(p + 1)\lambda D}{d}$$

This is the position of **(p+1)** bright fringe w.r.t. point O.

The distance between two successive bright fringes is called **fringe width** and is given by

$$\beta = x_{p+1} - x_p = \frac{(p + 1)\lambda D}{d} - \frac{p\lambda D}{d}$$

$$\beta = \frac{\lambda D}{d} \dots\dots\dots(2)$$

Minima or Dark fringes

If the path difference ($S_2P - S_1P$) = xd/D is an odd multiple of $\lambda/2$, then the point P will be the position of dark fringes or minima.

Thus for dark fringes,

$$\frac{xd}{D} = (2p + 1) \frac{\lambda}{2}$$

$$x = \frac{(2p + 1)\lambda D}{2d} \dots\dots\dots(3)$$

Eqn. (3) gives the position of different dark fringes.

IF

$$p = 0, \quad x_1 = \frac{\lambda D}{2d}$$

This is the position of first dark fringe w.r.t. point O.

IF

$$p = 1, \quad x_2 = \frac{3\lambda D}{2d}$$

This is the position of second dark fringe w.r.t. point O.

IF

$$p = 2, \quad x_3 = \frac{5\lambda D}{2d}$$

This is the position of third dark fringe w.r.t. point O.

IF

$$p = p, \quad x_p = \frac{(2p + 1)\lambda D}{2d}$$

This is the position of **p**th dark fringe w.r.t. point O.

If

$$p = (p + 1), \quad x_{p+1} = \frac{(2p + 3)\lambda D}{2d}$$

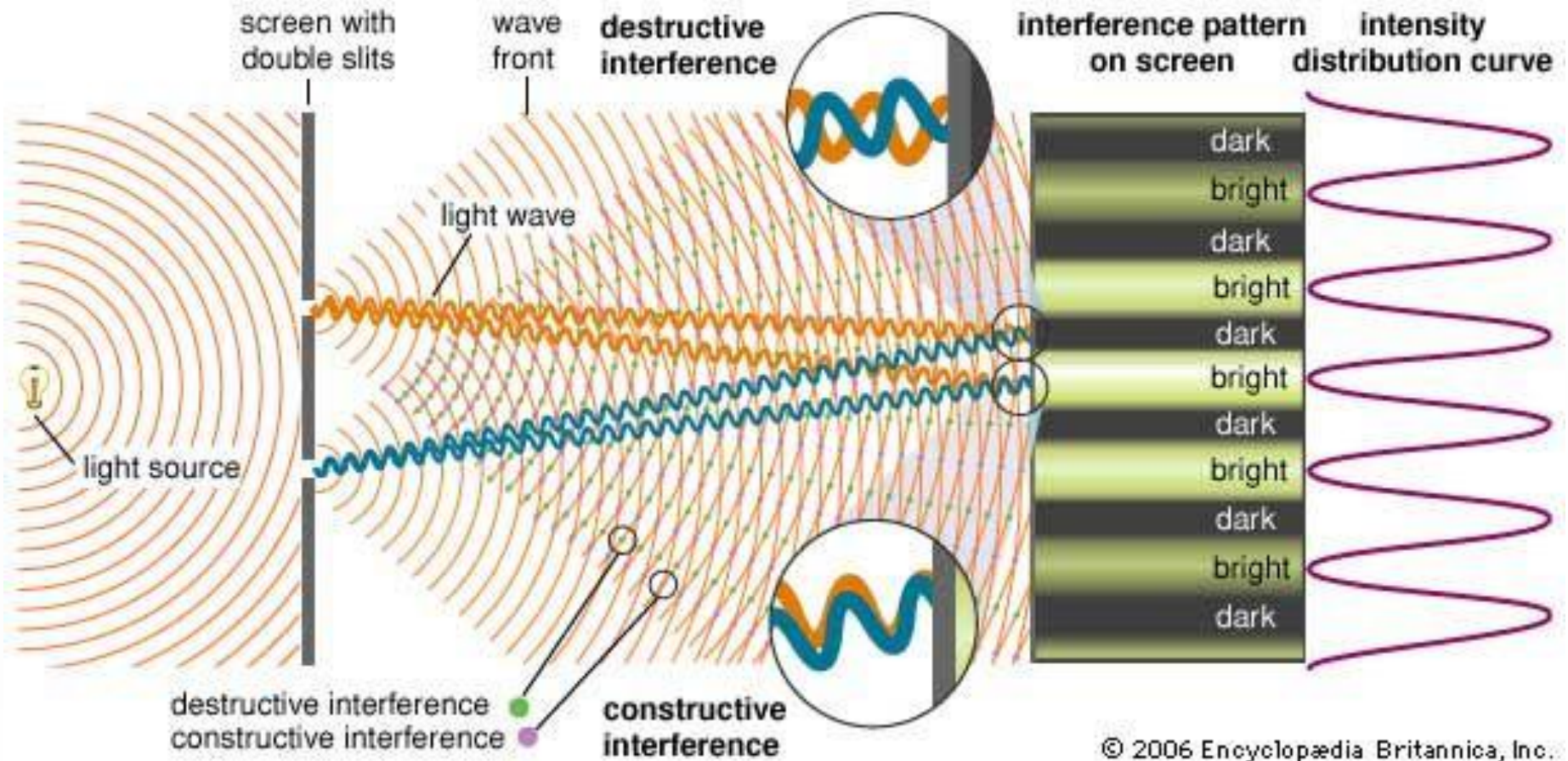
This is the position of **(p+1)** dark fringe w.r.t. point O.

The distance between two successive dark fringes is called fringe width (**β**) of the dark fringes which is given by

$$\beta = x_{p+1} - x_p = \frac{(2p + 3)\lambda D}{2d} - \frac{(2p + 1)\lambda D}{2d}$$

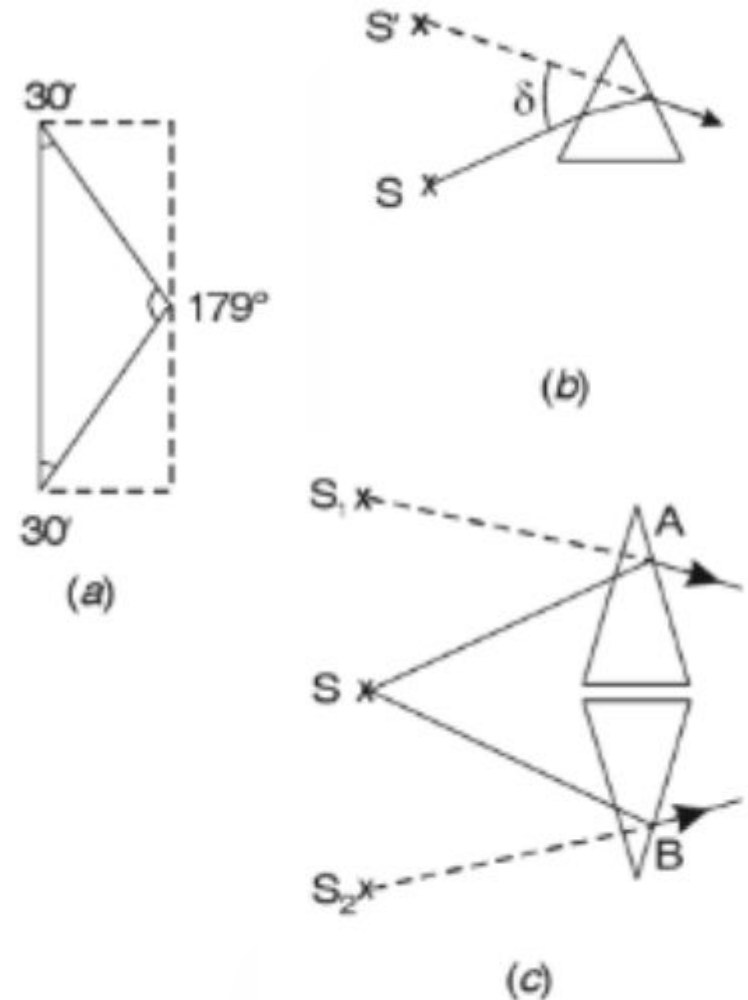
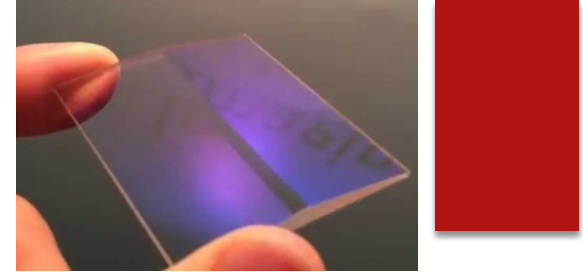
This eqn. (4), '**β=λD/d**' is called **Fringe width formula in Young's experiment**.

From eqns. (2) and (4), it is evident that the fringes width of bright fringe and dark fringe is the same.



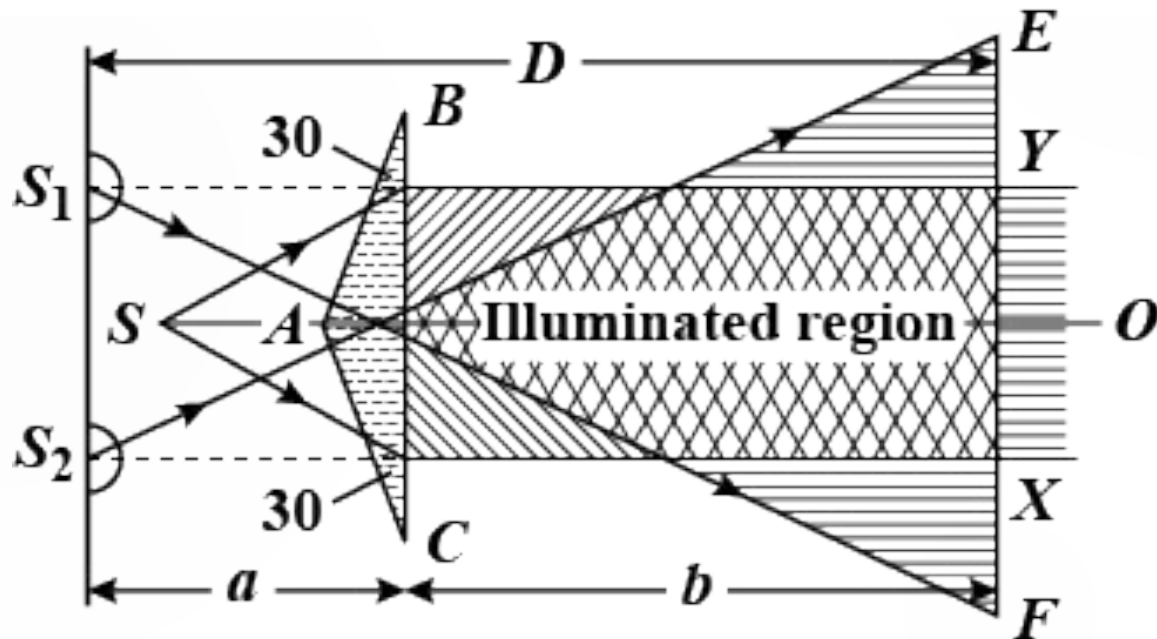
Fresnel's biprism

- Fresnel used a biprism to show interference phenomenon. The biprism consists of two prisms of very small refracting angles joined base to base. In practice, a thin glass plate is taken and one of its faces is ground and polished till a prism (Fig.a) is formed with an obtuse angle of about 179° and two side angles of the order of $30'$.
- When a light ray is incident on an ordinary prism, the ray is bent through an angle called the angle of deviation. As a result, the ray emerging out of the prism appears to have emanated from a source S' located at a small distance above the real source, as shown in Fig. (b).
- We say that the prism produced a virtual image of the source. A biprism, in the same way, creates two virtual sources S_1 and S_2 , as seen in Fig. (c). **These two virtual sources are images of the same source S produced by refraction and are hence coherent.**



Experimental Arrangement

- A monochromatic light source such as sodium vapour lamp illuminates a vertical slit S , acting as a narrow linear monochromatic light source.
- The biprism is placed in such a way that its refracting edge is parallel to the length of the slit S .
- A single cylindrical wavefront impinges on both prisms. One of these components is refracted from the upper portion of biprism and appears to come from S_1 whereas the other one is refracted through the lower portion and appears to come from S_2 .
- The virtual sources S_1 and S_2 are coherent and hence the light waves are in a position to interfere in the region beyond the biprism.
- If a screen is held there, interference fringes are seen.



Theory:

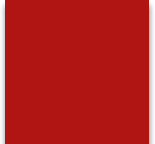
The theory of the interference and fringe formation in the case of Fresnel biprism is the same as described for the double-slit. As point O is equidistant from S1 and S2, the central bright fringe of maximum intensity occurs there. On both sides of O, alternate bright and dark fringes are produced. The width of the dark or bright fringe is given by eq.

$$\beta = \frac{\lambda D}{d} \dots\dots\dots(1)$$

Where ($D = a + b$) is the distance of the sources from the eyepiece. alternately

Determination of wavelength of light

The wavelength of the light can be determined using equation (1). For using the relation, the values of β , D and d are to be measured. These measurements are done as follows.



Let the separation between S1 and S2 be 'd' and the distance of slits and the screen from the biprism be a and b respectively i.e., $D = (a + b)$. If the angle of the prism is α and the refractive index is μ then

$$d = 2a(\mu - 1)\alpha$$

$$\begin{aligned}\lambda &= \beta d/D \\ &= \beta [2a(\mu - 1)\alpha] / a + b\end{aligned}$$

If a convex lens is mounted between the biprism and eyepiece. There will be two positions of the lens when the sharp images of coherent sources will be observed in the eyepiece. The separation of the images in the two positions are measured. Let these be d_1 and d_2 then the separation between the virtual source 'd' is given by

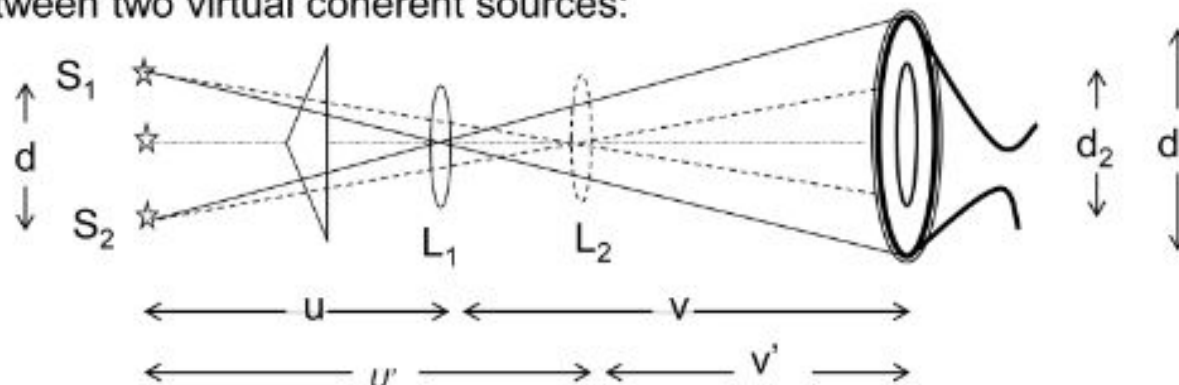
$$d = \sqrt{d_1 d_2}$$

$$\lambda = \beta d/D = \beta \sqrt{d_1 d_2} / [a + b]$$

$$\text{i.e., } \beta = [a + b] \lambda / 2a(\mu - 1)\alpha$$

(b) Determination of the distance between two virtual sources:

Displacement method is one of the methods to calculate the distance between two virtual coherent sources:



According to the linear magnification produced by the lens:

$$\frac{\text{size of the image (I)}}{\text{size of the object (O)}} = \frac{\text{distance between image and lens}}{\text{distance between object and lens}} \quad \frac{d_1}{d} = \frac{v}{u} \quad (1)$$

Further the lens moves towards the eyepiece and a focused image of virtual sources S_1 and S_2 is visible in eyepiece again. This time the image separation of S_1 and S_2 should be appear different (d_2) so that:

$$\frac{d_2}{d} = \frac{v'}{u'} \quad (2)$$

But $u = v'$ and $v = u'$

Thus equation (2) becomes

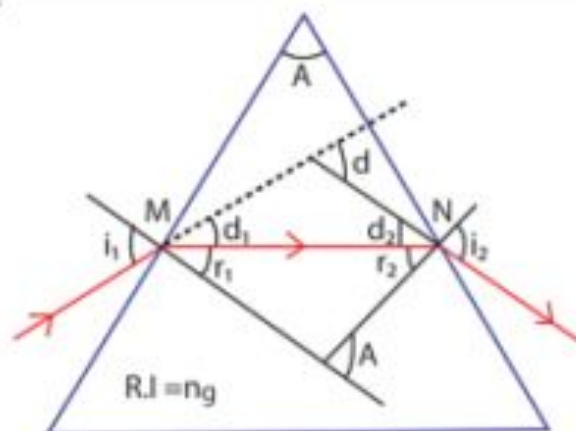
$$\frac{d_2}{d} = \frac{u}{v} \quad (3)$$

From equation (1) and (3), we get

$$\frac{d_2}{d} = \frac{d}{d_1}$$

or $d^2 = d_1 d_2$

$$d = \sqrt{d_1 d_2}$$



From the diagram, $d = d_1 + d_2$
 But $d_1 = i_1 - r_1$ and $d_2 = i_2 - r_2$
 $\Rightarrow d = (i_1 - r_1) + (i_2 - r_2)$

On simplifying $D = (i_1 + i_2) - (r_1 + r_2)$

But $r_1 + r_2 = A$

$\therefore D = i_1 + i_2 - A$ (a)

At M Snell's law becomes.

$$n_g \sin i_1 = n \sin r_1$$

For small angles this gives $i_1 = nr_1$ (b)

Similarly at N Snell's law becomes $i_2 = nr_2$ (c)

Substituting equation (b) and (c) in (a) gives

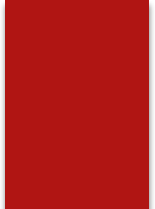
$$d = nr_1 + nr_2 - A$$

$$d = n(r_1 + r_2) - A$$

$$\text{but } r_1 + r_2 = A$$

$$\Rightarrow d = nA - A$$

$$\therefore d = (n - 1) A$$

- 
1. In Young's double-slit experiment, the slits are separated by 0.28 mm and the screen is placed 1.4 m away. The distance between the central bright fringe and the fourth bright fringe is measured to be 1.2 cm. Determine the wavelength of light used in the experiment.
 2. In Young's experiment, the distance between the two slits is 0.8 mm and the distance of the screen from the slits is 1.2m. If the fringe width is 0.75 mm, calculate the wavelength of light.
 3. In Young's experiment, the distance between the two images of the sources is 0.6 mm and the distance between the source and the screen is 1.5 m. Given that the overall separation between 20 fringes on the screen is 3 cm, calculate the wavelength of light used.
 4. Interference fringes are observed with a biprism of refracting angle 1° and refractive index 1.5 on a screen 100 cm away from it. The wavelength of light used is $5890 \text{ \AA}$. If the distance between the source and the biprism is 20 cm, the fringe width is?
 4. Calculate the fringe width in the pattern produced in a biprism experiment given that the wavelength of light employed is $6000 \text{ \AA}$, distance between sources is 1.2 mm and distance between the source and the screen is 100 cm. what will be the change in fringe width if the entire apparatus is immersed in water for which refractive index is $4/3$

